

## Machine Learning Fundamentals

Machine learning is a subset of artificial intelligence that enables systems to learn and improve from experience.

Supervised learning is the most common type of machine learning, where models learn from labeled training data.

Unsupervised learning deals with unlabeled data, finding hidden patterns or intrinsic structures. Clustering and association are common tasks.

Deep learning, a subset of machine learning, uses neural networks with multiple layers to progressively extract features from the input data.

The machine learning workflow typically involves data collection, preprocessing, model selection, training, evaluation, and deployment.